

CreditPulse AI

Decentralized Institutional Credit-Risk Oracle for RWA & Undercollateralized DeFi

Built for: **BUIDL CTC 2026 Fall — Creditcoin Hackathon** (DoraHacks)

Network: **Creditcoin CC3 Testnet** (Chain ID 102031)

Team: **Solo developer — Stepan Fedorov**

Project deck / whitepaper summary prepared for hackathon submission.

The Problem

Real-world-asset (RWA) financing and undercollateralized DeFi lending are two of the fastest-growing segments in on-chain finance — but both are held back by the same gap: there is no transparent, verifiable, on-chain way to assess a borrower's or counterparty's credit risk.

Traditional credit scoring is opaque, off-chain, and controlled by centralized bureaus. DeFi protocols that want to move beyond simple overcollateralized loans currently have no trustworthy, auditable risk signal to rely on — which limits institutional capital from entering on-chain credit markets at scale.

This creates three concrete gaps:

- **No verifiable risk data** — credit assessments happen off-chain, in black boxes, with no cryptographic guarantee of integrity.
- **No cross-chain trust** — risk data computed off-chain cannot be trustlessly proven and reused across chains.
- **No institutional-grade tooling** — existing DeFi risk tools are built for retail collateral ratios, not institutional credit underwriting.

The Solution

CreditPulse AI is a decentralized institutional credit-risk oracle that combines an AI-driven risk-scoring engine with an on-chain smart contract to generate, verify, and permanently record credit risk reports on Creditcoin CC3 Testnet.

Instead of trusting a centralized score, lenders and protocols can query CreditPulse AI's on-chain contract (CreditPulseASC) for a verifiable, tamper-evident risk report — generated by the AI scoring engine and anchored on-chain through cryptographic proofs, not just an API response.

How it works, end to end:

1. A borrower/asset profile is submitted through the CreditPulse AI web app.
2. The AI risk-scoring engine analyzes the profile and produces a credit risk report.
3. The report is proven via Creditcoin's cross-chain prover service and submitted on-chain through the Universal Smart Contract (USC) precompile.
4. The verified report is permanently recorded in the CreditPulseASC contract, publicly auditable on the Creditcoin explorer.

Architecture & Technology

Frontend & Application Layer

Next.js 16 (App Router) with React 19 and TypeScript. API routes implement the real application logic (risk scoring orchestration, on-chain submission), so the deployed app is a single full-stack Next.js service — no separate backend server is required in production.

Smart Contract Layer

The CreditPulseASC contract (Solidity 0.8.20, built and deployed with Hardhat) stores verified risk reports on-chain and exposes them for public, trustless querying.

Cross-Chain Verification — Universal Smart Contract (USC)

CreditPulse AI integrates Creditcoin's Universal Smart Contract (USC) precompile at address `0x0FD2`, which provides native cross-chain Merkle / continuity proof verification directly at the protocol level. The relayer calls the CC3 prover service to generate a proof for each risk report, then submits it through the USC precompile so the report's integrity is verified on-chain before being recorded in CreditPulseASC — removing the need for a trusted intermediary to vouch for the data.

Prover service:

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https://prover.cc3-testnet.creditcoin.network
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AI Risk Scoring Engine

An AI-driven scoring module evaluates borrower/asset inputs and produces a structured credit risk report, which is then anchored on-chain through the USC-verified pipeline described above.

Live Deployment & Links

Live application	https://frontend-gamma-pink-41.vercel.app
Smart contract (CreditPulseASC)	0x5BEC88F55ECA9038A9f03E77052314EfDC293Da5
Network	Creditcoin CC3 Testnet – Chain ID 102031
Block explorer	creditcoin-testnet.blockscout.com
Prover service	https://prover.cc3-testnet.creditcoin.network
GitHub repository	https://github.com/fedorov17808-svg/hakatons

Team

CreditPulse AI was designed, built, and deployed independently by a single developer — **Stepan Fedorov** — covering the full stack: smart contracts, on-chain/off-chain integration, the AI risk-scoring engine, and the frontend application, from architecture through production deployment on Creditcoin CC3 Testnet.